

GOIANIA, Brazil

WMO: 834240

Lat: 16.632S

Lon: 49.221W

Elev: 2450

StdP: 13.44

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	55.3	57.3	34.1	31.5	87.6	37.8	36.4	84.7	18.1	77.7	16.3	77.7	2.7	130	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	24.9	96.6	66.4	94.7	66.8	92.9	67.6	76.1	85.6	75.4	84.7	74.7	83.8	7.6	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.6	137.3	79.0	73.2	135.0	78.5	72.0	129.7	77.6	41.4	85.6	40.7	85.1	40.1	84.1	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.4	15.3	13.0	DB	49.1	99.3	3.0	2.5	46.9	101.2	45.2	102.6	43.5	104.0	41.4	105.9	(4)
(5)			WB	46.3	77.1	2.7	2.9	44.4	79.3	42.8	81.0	41.3	82.6	39.3	84.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.9	78.1	78.6	78.3	77.4	74.3	72.2	72.3	75.5	79.9	80.1	78.4	78.0	(6)	
	DBStd	3.77	2.48	2.36	2.26	2.41	2.67	2.77	3.08	3.01	3.08	3.28	2.82	2.38	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	5	0	0	0	0	1	2	2	0	0	0	0	0	(9)	
	CDD50	9822	870	800	879	822	753	667	692	791	896	932	852	868	(10)	
	CDD65	4353	405	380	414	372	289	220	229	326	446	467	402	403	(11)	
	CDH74	38646	2955	2884	3025	2993	2484	2153	2568	3858	5234	4663	3064	2765	(12)	
(13)	CDH80	16355	1133	1162	1155	1185	880	723	982	1828	2818	2296	1210	983	(13)	
(14)	Wind	WSAvg	5.2	5.2	4.8	4.5	4.6	4.5	4.8	5.4	6.2	6.4	5.8	5.4	5.4	(14)
(15) Precipitation	PrecAvg	61.9	10.0	9.3	10.0	4.2	1.1	0.3	0.1	0.4	1.9	5.1	8.7	10.7	(15)	
	PrecMax	75.1	19.1	15.4	20.7	9.6	3.0	2.8	0.6	2.0	4.8	11.6	12.3	18.7	(16)	
	PrecMin	42.8	3.2	5.1	3.7	0.0	0.0	0.0	0.0	0.0	0.1	0.9	4.6	6.2	(17)	
	PrecStd	8.0	3.9	2.5	3.5	2.7	0.9	0.6	0.1	0.6	1.3	2.8	2.2	3.5	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.4	93.6	92.9	92.8	90.0	89.4	91.1	95.1	98.7	98.9	95.4	92.7	(19)	
		MCWB	69.8	72.7	72.9	73.1	69.1	65.4	65.1	65.4	66.1	65.6	70.3	71.9	(20)	
	2%	DB	91.3	91.5	90.5	89.9	88.1	87.6	88.9	93.1	96.7	96.7	91.7	89.9	(21)	
		MCWB	70.3	72.4	72.5	71.2	68.8	66.2	63.9	64.4	65.7	66.2	70.7	71.3	(22)	
	5%	DB	89.3	89.5	88.7	88.1	86.4	86.0	87.6	91.2	94.9	93.7	89.5	88.0	(23)	
		MCWB	71.0	71.8	72.3	71.0	68.3	65.6	63.8	63.7	65.6	67.6	71.1	71.9	(24)	
	10%	DB	86.3	87.5	86.4	86.4	85.3	84.4	85.9	89.3	93.0	91.4	87.2	85.9	(25)	
		MCWB	71.9	71.9	72.4	71.0	67.9	65.3	63.6	63.1	65.4	68.3	71.1	72.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.4	77.3	76.6	76.0	74.4	71.0	69.4	70.5	74.0	75.3	75.8	76.3	(27)	
		MCDB	85.5	87.0	86.3	86.4	84.2	83.5	84.6	86.1	88.0	86.9	85.8	85.0	(28)	
	2%	WB	75.2	76.0	75.6	74.8	72.6	69.2	67.7	68.1	72.4	74.1	74.6	75.2	(29)	
		MCDB	84.2	85.1	84.9	84.9	82.6	81.4	83.2	83.2	85.9	85.3	84.0	83.9	(30)	
	5%	WB	74.4	75.0	74.8	74.0	71.3	68.1	66.3	66.4	71.2	73.1	73.8	74.4	(31)	
		MCDB	82.8	83.8	83.7	83.6	81.8	80.5	82.3	84.2	83.8	83.9	82.7	82.5	(32)	
	10%	WB	73.6	74.1	74.1	73.1	70.0	66.9	65.1	65.2	69.8	72.1	73.1	73.6	(33)	
		MCDB	81.5	82.3	82.3	82.2	79.6	79.3	80.9	83.9	81.3	82.4	81.6	81.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	17.5	16.9	19.1	21.9	24.7	26.6	27.1	24.9	20.5	17.1	15.8	(35)	
		MCDBR	21.5	21.2	20.3	21.4	24.2	27.1	29.4	30.3	29.3	25.8	21.4	19.8	(36)	
	5% WB	MCWBR	6.2	6.4	6.2	6.6	8.2	9.2	9.3	8.6	7.7	6.2	5.9	6.0	(37)	
		MCWBR	16.3	17.6	17.2	18.6	20.9	24.2	26.2	27.1	21.9	19.1	16.9	15.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.391	0.389	0.380	0.365	0.341	0.328	0.339	0.377	0.491	0.468	0.414	0.404	(40)	
		taud	2.450	2.447	2.489	2.502	2.539	2.574	2.500	2.360	2.056	2.186	2.364	2.404	(41)	
	Ebn at Noon		302	300	296	289	285	284	283	281	259	274	293	298	(42)	
		Edh at Noon	39	38	36	33	30	28	30	37	54	49	42	40	(43)	
(44) All-Sky Solar Radiation	RadAvg		1786	1829	1740	1722	1576	1507	1595	1806	1852	1848	1714	1742	(44)	
	RadStd		201	123	128	105	63	56	66	95	149	157	118	146	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page